SQL Practice Assignment – E-Commerce Data Analysis

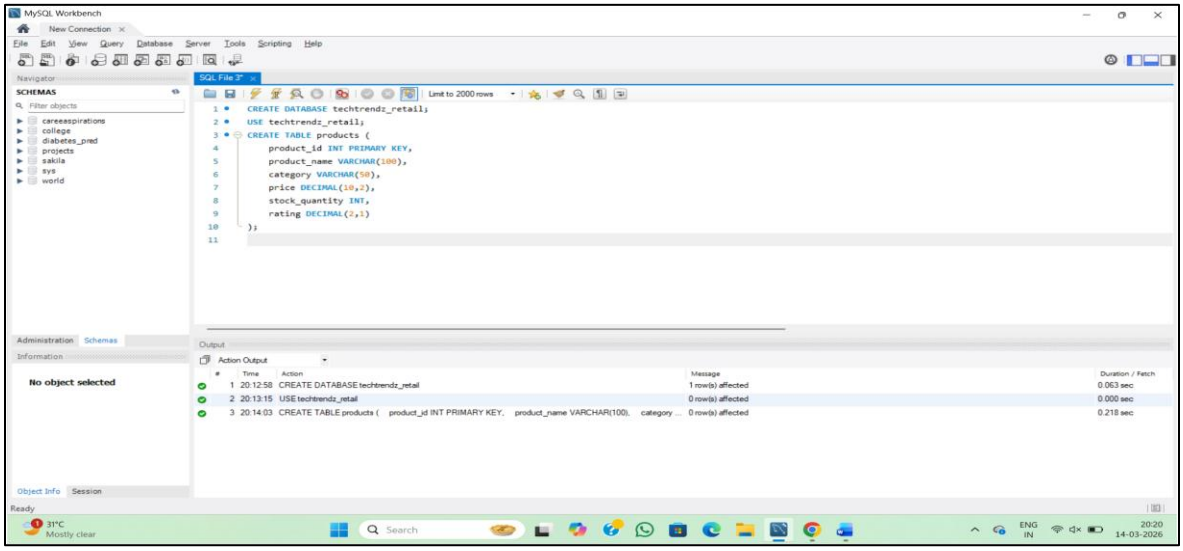
Step 1: Create Database

```
CREATE DATABASE techtrendz_retail;

USE techtrendz_retail;
```

Step 2: Create the Table

```
CREATE TABLE products (
   product_id INT PRIMARY KEY,
   product_name VARCHAR(100),
   category VARCHAR(50),
   price DECIMAL(10,2),
   stock_quantity INT,
   rating DECIMAL(2,1));
```



Step 3: Insert the Data

```
-- Insert product records into the products table

INSERT INTO products VALUES

(1,'Wireless Noise-Canceling Headphones','Electronics',299.99,50,4.8),

(2,'Ergonomic Office Chair','Furniture',150.00,20,4.5),

(3,'Gaming Laptop Pro','Electronics',1200.00,10,4.9),

(4,'Stainless Steel Water Bottle','Accessories',25.00,100,4.2),

(5,'Bluetooth Speaker Mini','Electronics',45.99,150,3.8),

(6,'Running Shoes - Size 10','Apparel',89.99,30,4.6),

(7,'Leather Wallet','Accessories',45.00,60,4.1),

(8,'4K Smart TV 55 inch','Electronics',600.00,15,4.7),

(9,'Yoga Mat','Fitness',20.00,200,4.3),

(10,'Coffee Maker','Appliances',75.50,40,4.4),

(11,'Smart Watch Series 5','Electronics',250.00,25,4.6),

(12,'Cotton T-Shirt','Apparel',15.00,300,3.9),

(13,'Gaming Mouse','Electronics',50.00,0,4.5),

(14,'Desk Lamp','Furniture',35.00,0,4.0),

(15,'Ceramic Mug Set','Kitchen',25.00,80,4.2);
```

SQL File 3*

```sql
INSERT INTO products VALUES
(1,'Wireless Noise-Canceling Headphones','Electronics',299.99,50,4.8),
(2,'Ergonomic Office Chair','Furniture',150.00,20,4.5),
(3,'Gaming Laptop Pro','Electronics',1200.00,10,4.9),
(4,'Stainless Steel Water Bottle','Accessories',25.00,100,4.2),
(5,'Bluetooth Speaker Mini','Electronics',45.99,150,3.8),
(6,'Running Shoes - Size 10','Apparel',89.99,30,4.6),
(7,'Leather Wallet','Accessories',45.00,60,4.1),
(8,'4K Smart TV 55 inch','Electronics',600.00,15,4.7),
(9,'Yoga Mat','Fitness',20.00,200,4.3),
(10,'Coffee Maker','Appliances',75.50,40,4.4),
(11,'Smart Watch Series 5','Electronics',250.00,25,4.6),
(12,'Cotton T-Shirt','Apparel',15.00,300,3.9),
(13,'Gaming Mouse','Electronics',50.00,0,4.5),
(14,'Desk Lamp','Furniture',35.00,0,4.0),
(15,'Ceramic Mug Set','Kitchen',25.00,80,4.2);
```

Output — Action Output

#	Time	Action	Message
1	20:12:58	CREATE DATABASE techtrendz_retail	1 row(s) affected
2	20:13:15	USE techtrendz_retail	0 row(s) affected
3	20:14:03	CREATE TABLE products (product_id INT PRIMARY KEY, product_name VARCHAR(100), category ...	0 row(s) affected
4	20:23:41	INSERT INTO products VALUES (1,'Wireless Noise-Canceling Headphones','Electronics',299.99,50,4.8), (2,'E...	15 row(s) affected Records: 15 Duplicates: 0 Warnings: 0

Part 2: Data Analysis Questions

A. Basic Data Retrieval (Select)

1. **Full Inventory Audit:** Select all columns and all rows from the table to verify your data insertion.

-- Display all columns and all rows from the products table

2. SELECT * FROM products;

SQL File 3*

```sql
(14,'Desk Lamp','Furniture',35.00,0,4.0),
(15,'Ceramic Mug Set','Kitchen',25.00,80,4.2);

-- Display all columns and all rows from the products table
SELECT * FROM products;
```

Result Grid

product_id	product_name	category	price	stock_quantity	rating
1	Wireless Noise-Canceling Headphones	Electronics	299.99	50	4.8
2	Ergonomic Office Chair	Furniture	150.00	20	4.5
3	Gaming Laptop Pro	Electronics	1200.00	10	4.9
4	Stainless Steel Water Bottle	Accessories	25.00	100	4.2
5	Bluetooth Speaker Mini	Electronics	45.99	150	3.8
6	Running Shoes - Size 10	Apparel	89.99	30	4.6
7	Leather Wallet	Accessories	45.00	60	4.1
8	4K Smart TV 55 inch	Electronics	600.00	15	4.7
9	Yoga Mat	Fitness	20.00	200	4.3
10	Coffee Maker	Appliances	75.50	40	4.4
11	Smart Watch Series 5	Electronics	250.00	25	4.6
12	Cotton T-Shirt	Apparel	15.00	300	3.9
13	Gaming Mouse	Electronics	50.00	0	4.5
14	Desk Lamp	Furniture	35.00	0	4.0
15	Ceramic Mug Set	Kitchen	25.00	80	4.2

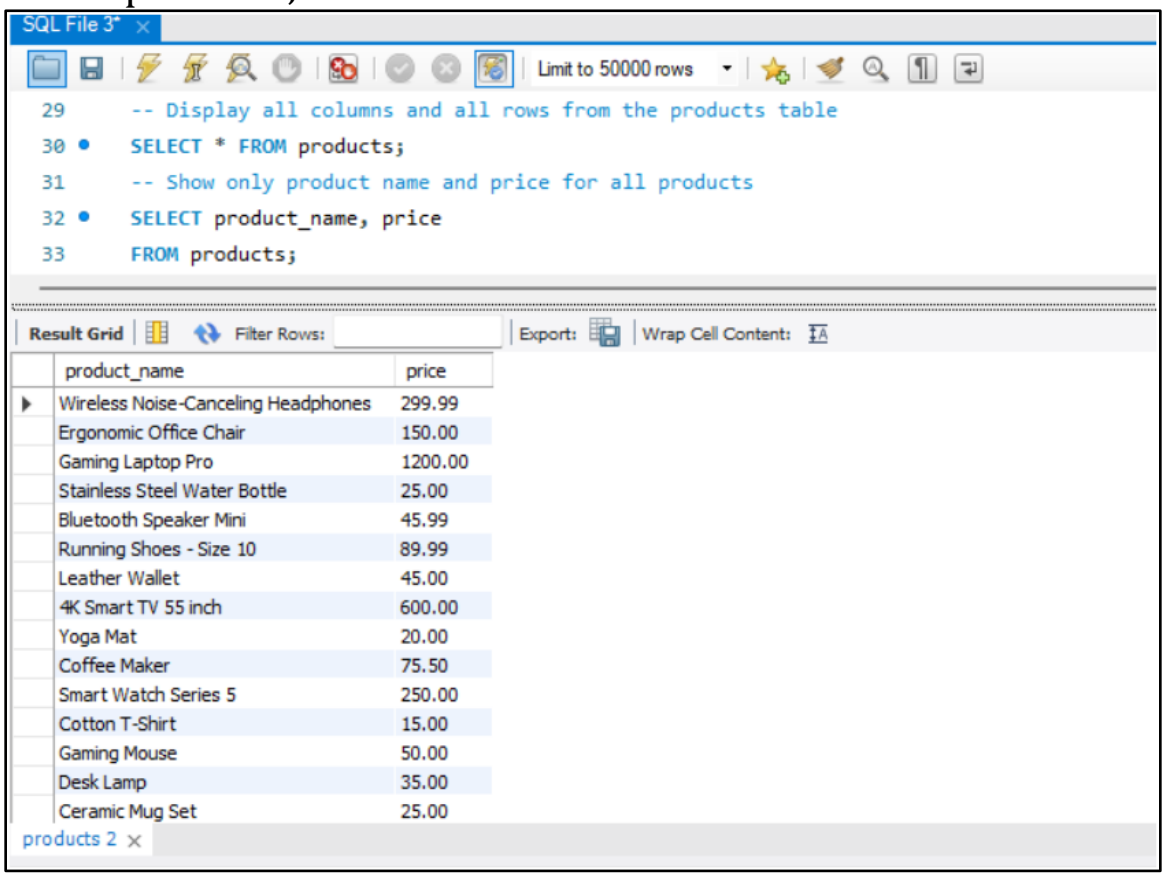
products 1

2. Price List: Select only the product_name and price columns for all products.

-- Show only product name and price for all products

SELECT product_name, price

FROM products;



product_name	price
Wireless Noise-Canceling Headphones	299.99
Ergonomic Office Chair	150.00
Gaming Laptop Pro	1200.00
Stainless Steel Water Bottle	25.00
Bluetooth Speaker Mini	45.99
Running Shoes - Size 10	89.99
Leather Wallet	45.00
4K Smart TV 55 inch	600.00
Yoga Mat	20.00
Coffee Maker	75.50
Smart Watch Series 5	250.00
Cotton T-Shirt	15.00
Gaming Mouse	50.00
Desk Lamp	35.00
Ceramic Mug Set	25.00

B. Filtering Data

3. High-Ticket Items: Select the product_name, category, and price for items that cost more than $200

-- Find products that cost more than $200

SELECT product_name, category, price

FROM products

WHERE price > 200;

```
-- Find products that cost more than $200
SELECT product_name, category, price
FROM products
WHERE price > 200;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

product_name	category	price
Wireless Noise-Canceling Headphones	Electronics	299.99
Gaming Laptop Pro	Electronics	1200.00
4K Smart TV 55 inch	Electronics	600.00
Smart Watch Series 5	Electronics	250.00

4. Category Focus: Select all columns for products where the category is specifically 'Electronics'.

-- Display all products that belong to the Electronics category

SELECT *FROM products

WHERE category = 'Electronics';

```
-- Display all products that belong to the Electronics category

SELECT *FROM products
WHERE category = 'Electronics';
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

product_id	product_name	category	price	stock_quantity	rating
1	Wireless Noise-Canceling Headphones	Electronics	299.99	50	4.8
3	Gaming Laptop Pro	Electronics	1200.00	10	4.9
5	Bluetooth Speaker Mini	Electronics	45.99	150	3.8
8	4K Smart TV 55 inch	Electronics	600.00	15	4.7
11	Smart Watch Series 5	Electronics	250.00	25	4.6
13	Gaming Mouse	Electronics	50.00	0	4.5

5. Low Stock Alert: Select the product_name and stock_quantity for items where the stock is less than 30.

-- Find products where stock quantity is less than 30

SELECT product_name, stock_quantity

FROM products

WHERE stock_quantity < 30;

SQL File 3*

Limit to 50000 rows

```sql
-- Find products where stock quantity is less than 30
SELECT product_name, stock_quantity
FROM products
WHERE stock_quantity < 30;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

product_name	stock_quantity
Ergonomic Office Chair	20
Gaming Laptop Pro	10
4K Smart TV 55 inch	15
Smart Watch Series 5	25
Gaming Mouse	0
Desk Lamp	0

6. Out of Stock: Find all columns for products that are currently completely out of stock (stock quantity is 0).

-- Display products that have zero stock

SELECT *

FROM products

WHERE stock_quantity = 0;

```sql
-- Display products that have zero stock
SELECT *
FROM products
WHERE stock_quantity = 0;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

product_id	product_name	category	price	stock_quantity	rating
13	Gaming Mouse	Electronics	50.00	0	4.5
14	Desk Lamp	Furniture	35.00	0	4.0
NULL	NULL	NULL	NULL	NULL	NULL

B. Sorting Data (ORDER BY)

7. Budget Friendly: Select product_name and price, sorting them by price in ascending order (cheapest first).

-- Show products sorted by price in ascending order (cheapest first)

SELECT product_name, price

FROM products

ORDER BY price ASC;

```
-- Show products sorted by price in ascending order (cheapest first)
SELECT product_name, price
FROM products
ORDER BY price ASC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

product_name	price
Cotton T-Shirt	15.00
Yoga Mat	20.00
Stainless Steel Water Bottle	25.00
Ceramic Mug Set	25.00
Desk Lamp	35.00
Leather Wallet	45.00
Bluetooth Speaker Mini	45.99
Gaming Mouse	50.00

8. Top Rated: Select product_name, category, and rating. Sort the results so the highest-rated products appear at the top (descending order).

-- Display products sorted by highest rating

SELECT product_name, category, rating

FROM products

ORDER BY rating DESC;

```
-- Display products sorted by highest rating
SELECT product_name, category, rating
FROM products
ORDER BY rating DESC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

product_name	category	rating
Gaming Laptop Pro	Electronics	4.9
Wireless Noise-Canceling Headphones	Electronics	4.8
4K Smart TV 55 inch	Electronics	4.7
Running Shoes - Size 10	Apparel	4.6
Smart Watch Series 5	Electronics	4.6
Ergonomic Office Chair	Furniture	4.5
Gaming Mouse	Electronics	4.5
Coffee Maker	Appliances	4.4

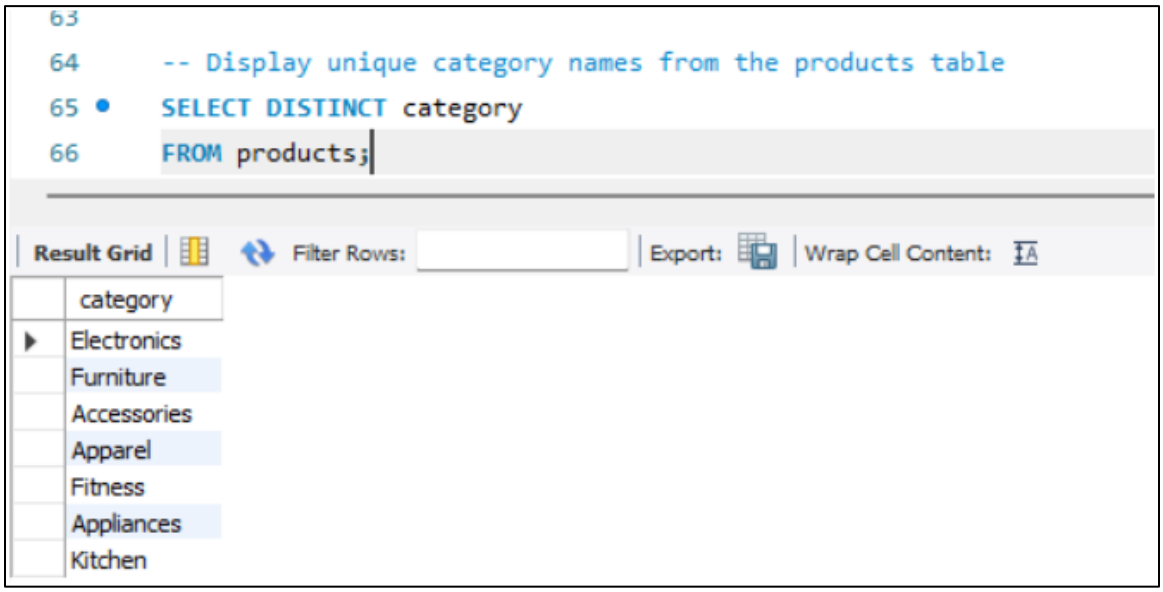
C.Unique Values (DISTINCT)Category List

9. Category List: Write a query to list all unique/distinct category names found in the table

-- Display unique category names from the products table

SELECT DISTINCT category

FROM products;



E. Complex Logic (Multiple Clauses)

10. Priority Restock: Select product_name, stock_quantity, and category for products in the 'Electronics' category that have a stock quantity less than 50. Sort the result by stock quantity (lowest stock first).

-- Find Electronics products with stock less than 50

-- Results are sorted by lowest stock first

SELECT product_name, stock_quantity, category

FROM products

WHERE category = 'Electronics'

AND stock_quantity < 50

ORDER BY stock_quantity ASC;

```sql
-- Find Electronics products with stock less than 50
-- Results are sorted by lowest stock first
SELECT product_name, stock_quantity, category
FROM products
WHERE category = 'Electronics'
AND stock_quantity < 50
ORDER BY stock_quantity ASC;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

product_name	stock_quantity	category
Gaming Mouse	0	Electronics
Gaming Laptop Pro	10	Electronics
4K Smart TV 55 inch	15	Electronics
Smart Watch Series 5	25	Electronics